



The Health On the Net Code of Conduct for medical and health Websites

C. Boyer^{a,*}, M. Selby^a, J.-R. Scherrer^{a,b}, R.D. Appel^{a,b}

^a*Health On the Net Foundation, DIM-24, rue Micheli-du-Crest, 1211 Geneva 14, Switzerland*

^b*Division of Medical Informatics, Geneva University Hospital, Geneva, Switzerland*

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Abstract

Internet has become one of the most used communication media. This and the fact that no constraining information publishing policy exists have created an urgent need to control the quality of information circulating through this media. To this purpose, the Health On the Net Foundation has initiated the Code of Conduct (HONcode) for the health/medical domain. This initiative proposes guidelines to information providers, with the aim, on the one hand, of raising the quality of data available on the Net and, on the other hand, of helping to identify Internet sites that are maintained by qualified people and contain reliable data. The HONcode mainly includes the following ethical aspects: the author's credentials, the date of the last modification with respect to clinical documents, confidentiality of data, source data reference, funding and the advertising policy. This article presents the HONcode and its evolution since it was launched in 1996. © 1998 Elsevier Science Ltd. All rights reserved.

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1. Introduction

The advent of the Internet, and especially in recent years the explosion of the World Wide Web (WWW or Web), has provided professionals, but also the general public, with the access to a wealth of information. In the medical and health domain, data and information available on the Web offer to individuals (and patients in particular) the possibility to investigate nearly

* Corresponding author. Tel.: +41-22-372-6273; Fax: +41-22-372-6198; E-mail: celia.boyer@hon.ch.

any topic of interest and to compare data from many different sources. Medical professionals can also use this media to obtain up-to-date information, or to exchange data and knowledge.

With the wide availability of Web server software it becomes increasingly easy to set up a Website and to publish any kind of data. Similarly, using Web browsers such as Netscape Navigator[™] or Microsoft Internet Explorer[™], most people can surf the Net and have access to these data. The problem is therefore no longer to find information, but to assess the credibility of the data source, as well as the relevance and correctness of a document retrieved from the Net. In many cases, no appropriate documentation is provided regarding the scientific design of a medical study, nor are studies made available that support given claims. The Code of Conduct for medical Websites (HONcode) [1] has been elaborated by the Health On the Net Foundation [2] to help unify and standardize the quality of medical and health information available on the WWW.

The first published version of the Code of Conduct, version 1.5 [3], was made available in July 1996 in response to concerns expressed by a number of people involved in medicine and health care, regarding the difficulty to determine the reliability of data on the Net. Of specific concern were the validity of research data, the quality of medical advice being given on certain Websites, as well as the lack of scientific data supporting claims related to commercially available treatments [4]. Also, Webmasters do not always include all the necessary useful information (source of document, e-mail address, date of last update, ...) on their Website. These omissions are in most cases free of malicious intent. Even more, discussion with these Webmasters showed that they were seeking guidelines for content presentation and were willing to improve their site by following some basic rules. The HONcode was therefore developed as a self-governance initiative that webmasters can accept or reject. By displaying the HONcode logo [5] on a Website, the owners of the site are demonstrating their desire to provide medical/health information, services or resources via the WWW that are of genuine value to the Net community and that meet specific quality standards. We hope that this approach will forestall draconian attempts at regulation by possibly inappropriate bodies.

Also, it was indicated that the HONcode would evolve according to the Net community demand. To meet requirements of the users and of the health and medical sites' webmasters, a new version of the Code of Conduct, version 1.6 was proposed [1].

In Section 2 we detail the principles and the methods of evaluation of the HONcode. Then, in Section 3 we describe the HONcode status in 1997 and finally, in Section 4, we present the world-wide use of this initiative and its impact on the Net community.

2. Materials and methods

A distinguished group of experts in medical informatics and telemedicine met to discuss the opportunities and risks associated with the Web at an international working conference in Geneva, Switzerland in September 1995 [6, 7]. Organisations represented included the World-Wide Web Consortium (W3C), European Community DGXIII, National Library of Medicine (NLM), World Health Organization (WHO), International Telecommunication Union (ITU), European Bioinformatics Institute (EBI), International Medical Informatics Association (IMIA), European Laboratory for Particle Physics (CERN) and the Japanese Ministry of

Health and Welfare. One of the pressing issues to emerge from the conference was the quality of information being made available on the Web. Post-conference analysis indicated further cause for concern:

- 113% compound annual growth rate of Internet users [8].
- Evidence that recently diagnosed patients were seeking medical information and details of potential treatments¹.
- Recognition that some patients were seeking information while in a state of stress/anxiety.
- The possibility of medical advice being offered by individuals lacking formal medical training and qualifications.
- Newly emerging applications and technologies providing significant information on individuals accessing Web servers.
- Ignorance of these technologies by most Web server visitors.

Physicians indicated the risk of patients pursuing unproven treatments rather than following recognised treatment/procedures. The authors also recognised the risk of patients seeking information on ‘sensitive’ medical topics, e.g. sexually transmitted diseases, being reluctant to do so if they could not be offered some guarantee of anonymity.

All of these observations pointed to the need for a clearly stated procedure/advisory to ensure that patients and medical professionals could have confidence in the medical/health information they were accessing on the Net and be offered personal confidentiality and anonymity where appropriate.

The HONcode version 1.6 contains a set of principles that owners of Web servers are invited to abide by. If they so agree, they may display the HONcode logo on their Web server. This logo must be linked to the HONcode’s Web page. The HONcode logo can be downloaded by webmasters for their site from the HON Website [5] and they must advise the Health On the Net Foundation by e-mail (HONcode@hon.ch) of their participation to this initiative.

3. The HONcode

The eight principles of HONcode version 1.6 are as follows:

1. Any medical advice provided and hosted on this site will only be given by medically trained and qualified professionals unless a clear statement is made that a piece of advice offered is from a non-medically qualified individual/organisation.
2. The information provided on this site is designed to support, not replace, the relationship that exists between a patient/site visitor and his/her existing physician.
3. Confidentiality of data relating to individual patients and visitors to a medical Website, including their identity, is respected by this Website. The Website owners undertake to honour or exceed the legal requirements of medical information privacy that apply in the country and state where the Website and mirror sites are located.

¹ ● An example is Dr. Andy Groves search for prostate cancer information (see Ref. [9]).

4. Where appropriate, information contained on this site will be supported by clear references to source data and, where possible, have specific HTML links to that data. The date when a clinical page was last modified will be clearly displayed (e.g. at the bottom of the page).
5. Any claims relating to the benefits/performance of a specific treatment, commercial product or service will be supported by appropriate, balanced evidence in the manner outlined in principle (4).
6. The designers of this Website will seek to provide information in the clearest possible manner and provide contact addresses for visitors that seek further information or support. The web-master will display his/her e-mail address clearly throughout the Website.
7. Support for this Website will be clearly identified, including the identities of commercial and non-commercial organisations that have contributed funding, services or material for the site.
8. If advertising is a source of funding it will be clearly stated. A brief description of the advertising policy adopted by the Website owners will be displayed on the site. Advertising and other promotional material will be presented to viewers in a manner and context that facilitates differentiation between it and the original material created by the institution operating the site.

The Health On the Net Foundation reviews sites bearing the HONcode logo on a regular basis and uses a variety of search techniques to identify Web servers using the logo that have not been registered with the HON Foundation.

In the event that a Web server is blatantly misusing the HONcode, a clear procedure exists for notifying, then warning the owner of that Web server. Continued misuse will result in an advisory notice being issued to all HONcode initiative participants, the Hospital-Webmaster mailing list [10] and other appropriate channels. Legal action for trademark and copyright infringement will be initiated as a last resort.

3.1. Process to actively support the HONcode principles

For webmasters wishing to support this initiative and adhere to the HONcode principles, a short questionnaire has been created to verify if the Website follows all principles or to identify modifications that could possibly be required [11]. This questionnaire provides the option to abide by the HONcode principles, by receiving the corresponding HTML code by e-mail. This code includes the default HONcode logo linked to the HONcode's Web page, that can simply be copied and pasted into a site's HTML document. It will thus automatically appear on that Website. Websites supporting the HONcode initiative must display the following statement:



We subscribe to the HONcode principles
of the Health On the Net Foundation

To illustrate the registration process, let us assume a new site has decided to adhere to the principles. The site's webmaster has filled the questionnaire. The outcome allows the Website manager to identify that principle (7) is not followed. It requests the required modification in

order to receive validation of the site. The webmaster is thus committing herself to provide the modification within a reasonable time period. Once this is done the Foundation team is aware of the adhesion to the principles. The site can then be registered in the MedHunt's database, the HON medical/health documents finder [12].

3.2. Its status in 1997

The principles are dynamic. After discussion and consensus between the Health On the Net Foundation and the owners of Web servers bearing the HONcode logo, modifications to the Code of Conduct, version 1.5 were discussed in an open manner.

On the 14th January 1997 a request for comments (RFC) was issued asking for feedback on the new version of the HON Code of Conduct. HON received considerable feedback from many individuals and institutions. These included a range of constructive comments, some of which, naturally, conflicted with each other. The proposed modifications were also discussed on several mailing lists, such as Hospital-Webmaster [10] and MEDWEBMASTERS-L [13]. After analysis and further discussions with many respondents, a new version of the Code of Conduct (version 1.6) was proposed.

3.3. Other initiatives

Other ongoing activities related to the development and assessment of website quality evaluation tools can be reported. A project at the University of Georgia is establishing criteria to evaluate the quality of Internet resources in general [14]. The audience for this project are mainly educators. A similar approach for the health domain has been initiated by the Health Information Technology Institute (HITI) [15], defining criteria to evaluate the quality of health information [16]. This is dedicated to Internet visitors to assess the relevance of health documents. Both initiatives are in the development stage. Other efforts have been undertaken in other areas, in order to protect the visitor. The most critical and sensible area, pornography, now restricts access to adult content. Most of these types of Websites warn the visitors, in particular teenagers under 18, with an agreement of statement, and the use of one of the current adult age verification services used to block the adult content pages [17].

4. Results

Since the launch of this initiative in July 1996, feedback from institutions, medical professionals and patients has been extremely positive. Those Web servers bearing the HONcode logo offer reassurance to visitors that the information being provided and the operating principles of the Web server meet specific quality standards. The authors believe that this initiative helps address a key prerequisite in realising the potential benefits to the medical and health communities of the Web and on-line services.

Participants in the HONcode initiative are growing daily. The desire from the Net community and pharmaceutical industry to participate in this initiative demonstrates the

growing need for information quality assessment procedures. At the present time there are 684 medical and health Web servers showing the HONcode logo.

5. Discussion and conclusions

World wide owners of medical/health Websites are aware of the need to unify the quality, to ensure that accurate, unbiased, ethical health/medical information prevail on the Internet. There is also a need to educate and continue providing clear guidance for medical/health Website managers. Additionally and from a legal perspective, it is apparent that:

- There is a lack of common law decisions on matters relating to medical information/practice on the Web, Internet and On-line Services.
- The ability of legislative bodies to address the dynamics of this global on-line medium in a timely manner is questionable.
- The Net community is displaying considerable resistance towards governmental regulation.

For these reasons, it has been decided that self-governance through a widely agreed set of principles could provide a timely, effective solution to the concerns outlined above, improving the quality and practice of medical/health information and advice on the Net.

As the Internet is evolving according to users needs, so will the HONcode. Further principles and/or modifications will be incorporated after consultations of all the contributing parties. Currently the HONcode policy consists of the registration into the MedHunt's database, along with a description of the Websites supporting this initiative and the use of tracking techniques applied to those sites. A future method of policy of this self governance initiative is to provide to each Website desiring to abide and follow the HONcode a unique random number associated with the validation of the questionnaire. A HONcode logo that is displayed without having been registered will launch a warning message. This method provides an automatic list of registered Websites abiding to the HONcode along with the Websites' URL and e-mail addresses.

The Health On the Net Code of Conduct (HONcode) was drafted with the help of physicians and respected members of the Net community in Europe and North America. It provides a series of principles that Health On Net and the many contributors to the Code around the World hope will help improve the quality and presentation of information to site visitors and respect the confidentiality of visitors seeking information or help.

The HONcode initiative has demonstrated that precise guidelines for improving the quality of information on the Web are requested both by webmasters and users. It has also shown that simple means such as a self-governing code of conduct can serve as a first step towards a high quality medical and health Web [18].

6. Summary

The growth in the number of medical and health Websites and the varying quality of medical and health information currently available on the World-Wide Web have created the need for guidelines to help homogenise the quality of such medical information. In view of the rapid proliferation of new medical and health Websites, it is often difficult for a user to identify sites offering useful and reliable information or advice. The Health On the Net Foundation (HON) has initiated the Code of Conduct (HONcode) in response to the above concerns. The Code of Conduct for medical Websites (HONcode) is an initiative to help unify the quality of medical and health information available on the WWW. This initiative offers information providers good practice guidelines and offers users an indication of commitment to quality by those providers. Websites abiding to the HONcode have to display the HONcode logo on their Website. The HONcode logo demonstrates the desire of the site's owner to provide medical/health information, services or resources via the WWW that are of genuine value to the Net community and that meet specific quality standards defined by the HONcode principles. The HONcode currently contains 8 principles that are observed by more than 500 medical Websites displaying the corresponding logo.

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Celia Boyer received her M.Sc. in Applied Physics from Université de Luminy in 1993 and an Engineer Diploma in Computer Science from Ecole Polytechnique Fédérale de Lausanne. She is the webmaster of the Health On the Net Foundation, Geneva Hospital.

Mark Selby is Executive director of the Health On the Net Foundation and President of the Internet Society Geneva Chapter. He has held senior executive position at Digital Equipment, Xerox and Océ. Appointed as an expert to the European Commission on Electronic Data Interchange in 1987, Mark Selby has worked extensively in the field of computer networking.

Professor Jean-Raoul Scherrer is a certified internist who received his M.D. from Geneva University Medical School in 1959. He became laureate of the Geneva Medical School for his doctorate thesis in 1965. From 1967 to 1969, he collaborated in research in physics at Brookhaven National Laboratory in New York. He became lecturer on medical informatics at the Geneva University Medical School in 1971 and a full Professor at the same medical school in 1979. He is Director of Medical Informatics at the Geneva University Hospital.

Ron D. Appel received his Ph.D. in Computer Science from Geneva University in 1997. After a post-doc at Harvard School of Public Health, and Brigham and Women's Hospital, he joined the Medical Informatics Division at Geneva University Hospitals where he is leading the Molecular Imaging and Bioinformatics Laboratory. Ron D. Appel is also member of the Executive Committee of the Health On the Net Foundation.